

Model Rocketry 2026 Preliminary Design Review (PDR) Outline

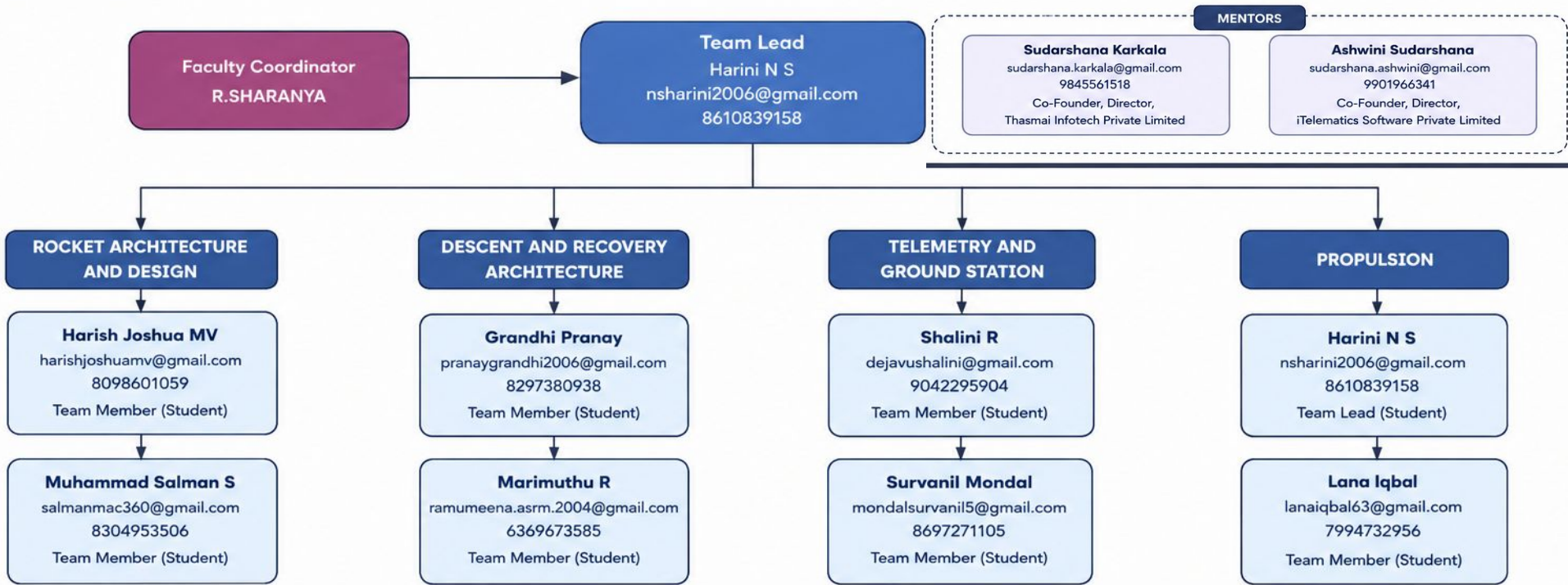
Team : 2026-INSPACe-ROCKETRY-059



PDR Overview

- Vehicle Architecture Baseline
- Propulsion Subsystem Design
- Recovery & Electrical Infrastructure
- Flight Simulations & Analytical Results
- Ground Station Operations & Compliance

Team Composition



Rocket configuration overview

ASTROFORGE VAYU is a one-stage sounding rocket designed to attain an altitude of 1000 ± 100 m and carry a 1 kg CAN-7USAT payload with an aerodynamically efficient configuration that can maintain flight stability, good structure, and recovery dependability.

The idea, which makes use of a Biomimetic Flow Management System (BFMS), was inspired by the fluid dynamic efficiency of the manta ray. By applying hydrodynamics from a manta ray's motion to an aerospace platform, this design increases flow attachment, decreases drag, and improves pressure recovery close to the fin-body intersection area.

This strategy can improve performance while adhering to competition regulations and maintaining structural simplicity and manufacturability.



Rocket configuration overview

Vehicle Designation	ASTROFORGE VAYU
Vehicle Type	Single-Stage Model Rocket
Mission Objective	Deployment of 1 kg CAN-7USAT Payload
Target Altitude	1000 ± 100 m
Total Length	1510 mm
Maximum Diameter	94 mm

Lift-Off Mass	6.6 kg
Recovery Method	Dual Deployment Recovery
Stability Margin	1.77 Calibers
Flight Regime	Subsonic
Fin Configuration	Four-Fin Swept Arrangement
Structural Material	PVC / PVC / PVC / PVC / PVC / Fiberglass composite



Rocket configuration overview

Biomimetic Flow Management Architecture

Engineering Principle:

Manta rays achieve remarkable efficiency in fluid environments due to:

- Smooth body-to-wing transitions
- Controlled vortex generation
- Efficient pressure recovery
- Reduced flow separation
- Stable directional control

1. Blended Fin-Root Transition System
2. Aft Flow-Recovery Fairing System
3. Swept Biomimetic Fin Geometry

These principles have been adapted into the rocket architecture through three major design elements.

Rocket configuration overview

Biomimetic Flow Management Architecture

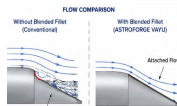
1. Blended Fin-Root Transition System

Traditional rockets suffer from localized turbulence and interference drag at the fin-body junction.

ASTROFORGE VAYU utilizes enlarged blended fillets that create a smooth aerodynamic transition between the airframe and fin surfaces.

Benefits

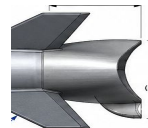
- Reduced interference drag
- Improved structural load transfer
- Increased airflow attachment
- Reduced boundary layer separation



2. Aft Flow-Recovery Fairing System

A pair of manta-inspired aerodynamic fairings are integrated ahead of the fin can.

These fairings gradually redirect airflow into the fin region and reduce abrupt pressure gradients.



Benefits

- Improved pressure recovery
- Reduced wake turbulence
- Improved aerodynamic efficiency
- Enhanced stability during ascent

Manta rays Inspiration

3. Swept Biomimetic Fin Geometry

The fin geometry emulates controlled vortex generation principles observed in manta ray propulsion.



Benefits

- Enhanced directional stability
- Reduced aerodynamic drag
- Improved damping characteristics
- Reduced crosswind sensitivity

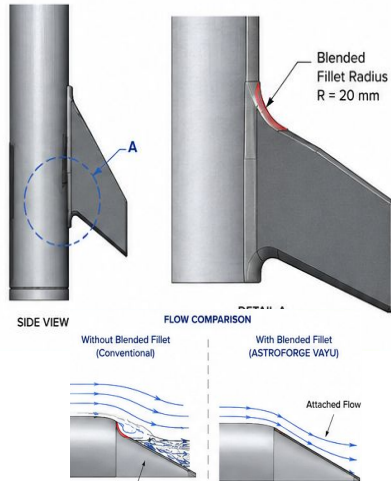
Rocket configuration overview

Biomimetic Flow Management Architecture

Manta rays Inspiration

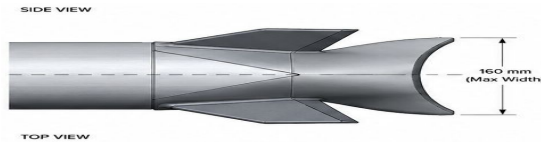
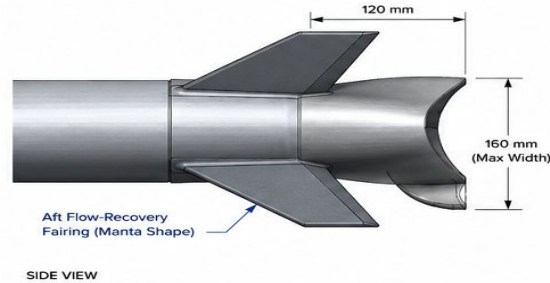
1. BLENDED FIN-ROOT TRANSITION SYSTEM

Smooth blended fillet between fin and body reduces flow separation, minimizes interference drag and improves structural load transfer.



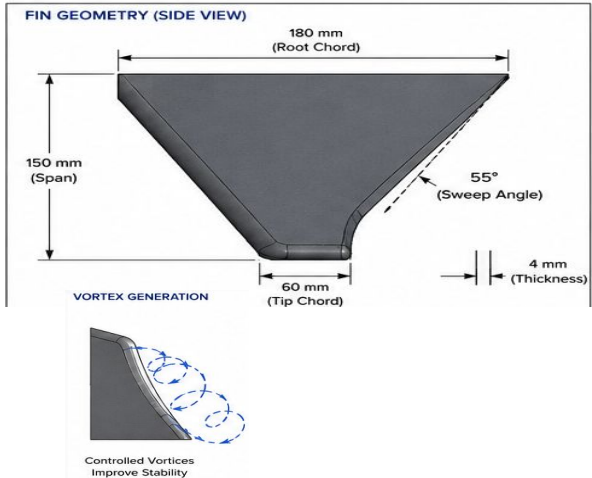
2. AFT FLOW-RECOVERY FAIRING SYSTEM

Aft fairings gradually guide the airflow into the fin region, recover pressure and reduce wake losses.



3. SWEEPED BIOMIMETIC FIN GEOMETRY

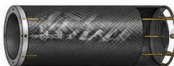
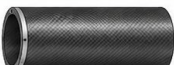
Swept fin geometry emulates controlled vortex generation as observed in manta ray propulsion for enhanced stability and reduced drag.





Rocket configuration overview

Structural Configuration

SECTION	LENGTH (mm)	MATERIAL	PRIMARY FUNCTION
 Nose Cone	280	PVC / Fiberglass Composite	 Aerodynamic shaping and recovery attachment
 Payload Bay	200	PVC / Fiberglass Composite	 Accommodation of 1 kg CAN-7USAT
 Avionics Bay	120	PVC / Fiberglass Composite	 Flight computer and sensor integration
 Recovery Bay	200	PVC / Fiberglass Composite	 Dual deployment recovery system
 Main Airframe	380	PVC / Fiberglass Composite	 Structural load carrying member
 Motor Bay	330	Fiberglass + PVC / Phenolic Mount	 Propulsion integration

Total Length =
1510 mm

Maximum
Diameter = 94
mm



Rocket configuration overview

Material Selection and Structural Justification

The material selection for ASTROFORGE VAYU was driven by four primary design objectives: structural integrity, weight reduction, manufacturability, and aerodynamic performance. Each material was selected based on the loading environment, functional requirements, and expected operational conditions experienced during launch, ascent, payload deployment, recovery, and landing.

Nose Cone

The nose cone is manufactured from PVC / Fiberglass composite due to its excellent stiffness-to-weight ratio and ability to maintain a precise aerodynamic profile. The fiberglass structure provides sufficient strength to withstand aerodynamic loading during ascent while minimizing forward mass accumulation that could adversely affect stability.

Payload Bay

The payload bay is constructed from PVC / Fiberglass composite tubing. This section houses the 1 kg CAN-7USAT payload and must provide adequate structural protection during launch acceleration, flight vibration, and recovery impact. Fiberglass offers high rigidity while remaining lightweight and corrosion resistant.

**Total Length =
1510 mm**

**Maximum
Diameter = 94
mm**

Rocket configuration overview

Total Length = 1510 mm

Maximum Diameter = 94 mm

Material Selection and Structural Justification

Avionics Bay

The avionics bay utilizes PVC / Fiberglass composite construction to provide a rigid enclosure for sensitive electronics including the flight computer, telemetry system, altimeters, GPS receiver, and sensors. The material also provides excellent dimensional stability, reducing the possibility of sensor misalignment during flight.

Recovery Bay

The recovery bay experiences significant transient loads during parachute deployment. PVC / Fiberglass composite was selected because of its ability to resist localized stress concentrations while maintaining low structural mass. The material provides a robust housing for parachutes, shock cords, and deployment mechanisms.

Main Airframe

The main airframe functions as the primary load-bearing structure of the vehicle. PVC / Fiberglass composite was selected because of its high bending stiffness, torsional rigidity, fatigue resistance, and low mass. These properties ensure that the vehicle remains structurally stable during acceleration, coast, and recovery phases.

Fin System

The fin assembly is fabricated from G10 fiberglass laminate. G10 offers exceptional strength, impact resistance, and dimensional stability. Since fins are subjected to significant aerodynamic forces and landing impacts, G10 provides a highly reliable solution while maintaining relatively low weight.



Rocket configuration overview

Total Length = 1510 mm

Maximum Diameter = 94 mm

Material Selection and Structural Justification

Bulkheads

Aircraft-grade birch plywood is used for bulkheads throughout the rocket. Bulkheads transfer axial loads between sections and provide mounting points for avionics, recovery hardware, and shock cord attachments. Birch plywood offers high compressive strength, excellent machinability, and proven performance in model rocketry applications.

Motor Mount Tube

The motor mount tube is manufactured from phenolic composite material. This section experiences elevated temperatures and direct propulsion loads. Phenolic composites exhibit excellent thermal resistance, dimensional stability, and structural stiffness, making them ideal for motor retention and thrust transfer.

Biomimetic Fairings

The manta-inspired flow-recovery fairings employ a foam-core and fiberglass-skin sandwich construction. The foam core allows complex aerodynamic contours to be fabricated while minimizing weight. The fiberglass skin provides the required structural strength and surface finish needed for aerodynamic performance.

Shock Cord System

The recovery harness utilizes Kevlar shock cords due to their exceptional tensile strength, heat resistance, and durability. Kevlar is capable of surviving the dynamic shock loads generated during parachute deployment while resisting heat from ejection gases and environmental degradation.

Structural Fasteners

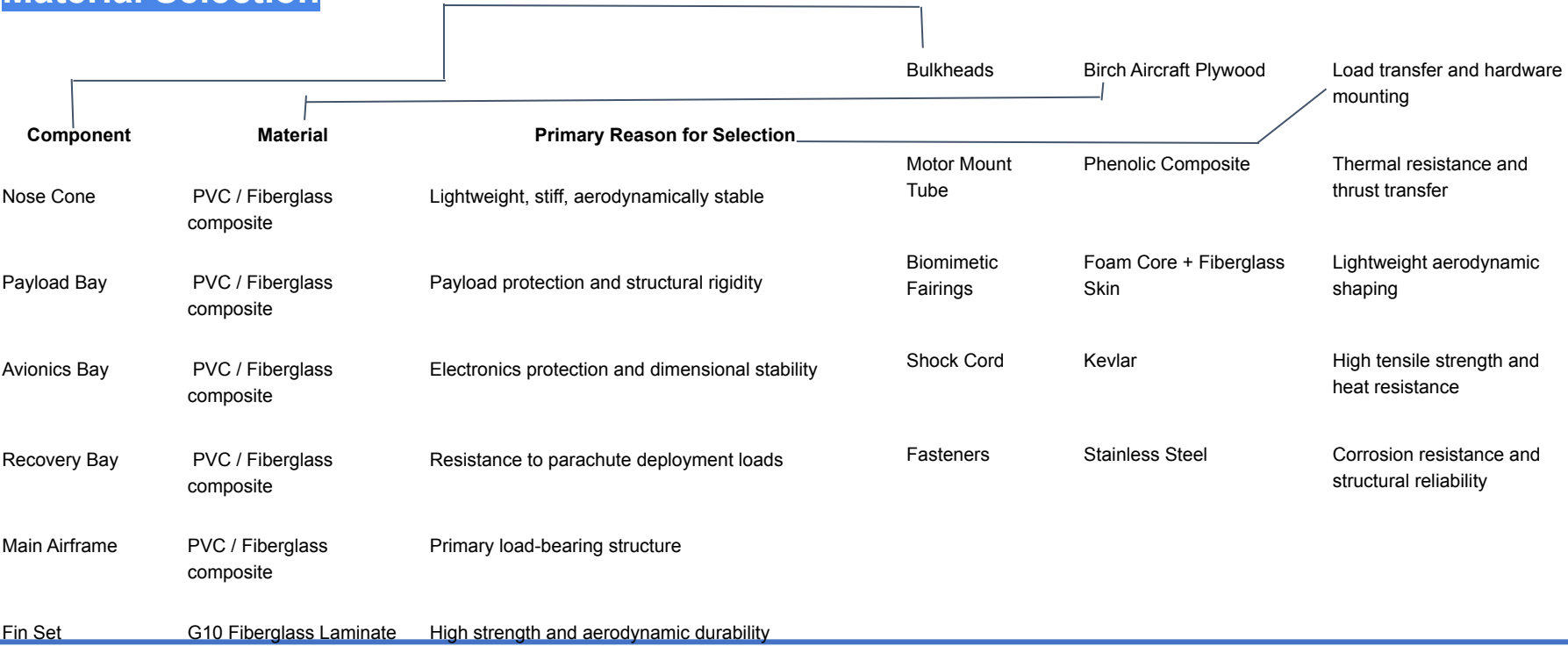
All mechanical fasteners are stainless steel. Stainless steel was selected for its corrosion resistance, fatigue performance, reliability, and ability to maintain structural integrity under repeated assembly and flight operations.



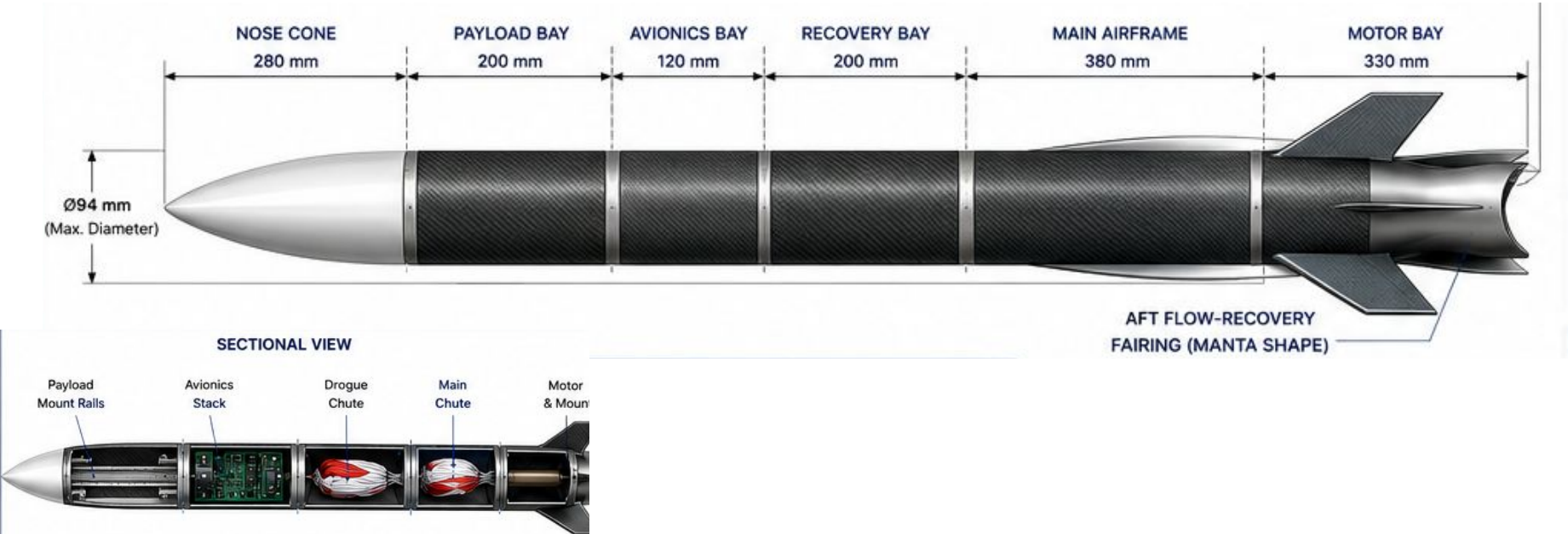


Material Selection

Rocket configuration overview



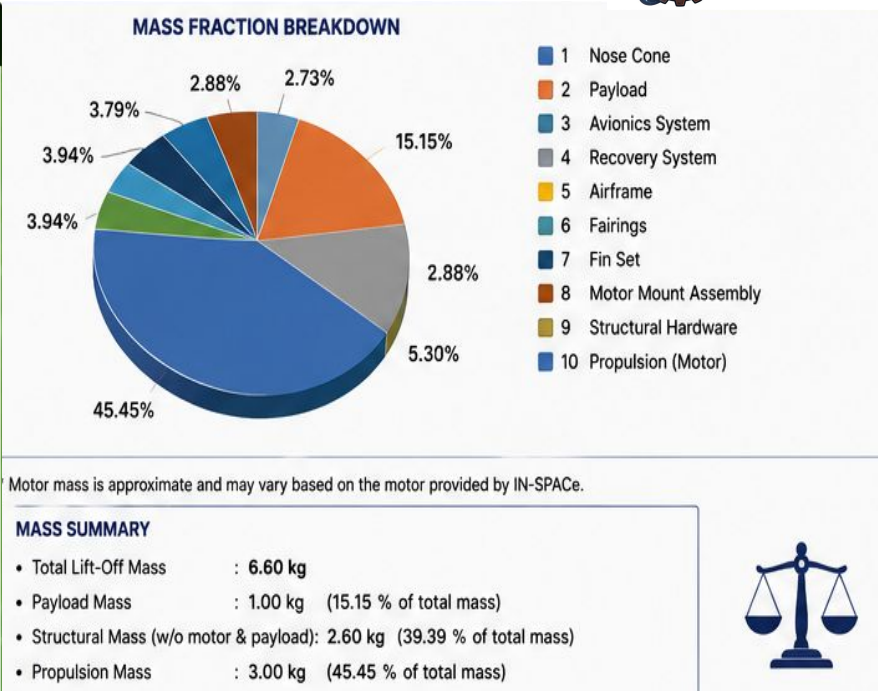
Rocket configuration overview



Rocket configuration overview



VEHICLE MASS BUDGET						
						Total Lift-Off Mass 6.60 kg
Sr. No.	Component	Qty.	Unit Mass (g)	Total Mass (g)	Mass Fraction (%)	Notes
1	Nose Cone (Fiberglass)	1	220	220	3.33	Includes bulkhead
2	Payload (CAN-7USAT)	1	1000	1000	15.15	Mission payload
3	Avionics System	1	350	350	5.30	Flight computer, sensors, telemetry, wiring
4	Recovery System	1	300	300	4.55	Parachutes, shock cord, deployment system
5	Airframe (Fiberglass Tubes)	1	850	850	12.88	All body sections
6	Biomimetic Fairings	2	180	180	2.73	Aft flow-recovery fairings (pair)
7	Fin Set (G10)	4	260	260	3.94	4 Fins with fillets
8	Motor Mount Assembly	1	250	250	3.79	Motor tube, centering rings, hardware
9	Structural Hardware	-	190	190	2.88	Fasteners, adhesives, retainers, etc.
10	Propulsion (Solid Motor)*	1	3000	3000	45.45	As per IN-SPACE provided motor (approx.)
TOTAL LIFT-OFF MASS				6,600 g	100 %	



Rocket configuration overview

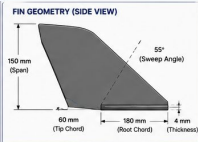
FIN SYSTEM CONFIGURATION

FIN CONFIGURATION OVERVIEW

- Configuration : 4 Fins
- Placement : Symmetrical (90° apart)
- Material : G10 Fiberglass Laminate
- Airfoil : NACA 0012 (Symmetric)
- Attachment : Epoxy Bonded + Mechanical Fasteners
- Fillet : 20 mm Radius (Blended)

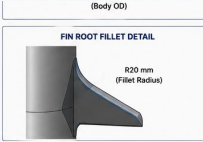
FIN PARAMETERS			
Parameter	Value	Parameter	Value
Number of Fins	4	Sweep Angle (Leading Edge)	55°
Root Chord	180 mm	Airfoil	NACA 0012
Tip Chord	60 mm	Fin Root Fillet Radius	20 mm
Span (Height)	150 mm	Fin Material	G10 Fiberglass Laminate
Thickness	4 mm	Fin Attachment	Epoxy + Mechanical Fasteners

FIN GEOMETRY (SIDE VIEW)

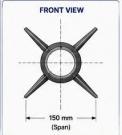


(Body OD)

FIN ROOT FILLET DETAIL



FRONT VIEW



ISOMETRIC VIEW



FIN DESIGN RATIONALE

- Sweep fins reduce drag and delay flow separation.
- G10 fiberglass provides high strength and impact resistance.
- Large root chord ensures structural strength.
- Blended fillet reduces interference drag and improves load distribution.

Rocket configuration overview

Design Justification

The selected geometry provides an optimum balance between:

- Static stability
- Structural stiffness
- Drag reduction
- Manufacturability

1. CENTER OF GRAVITY (CG) ANALYSIS

No.	Component	Mass (g)	CG Location from Nose Tip x_i (mm)	Moment ($m_i x_i$) (g-mm)
1	Nose Cone (Fiberglass)	220	140	30,800
2	Payload (CAN-7USAT)	1000	380	380,000
3	Avionics System	350	540	189,000
4	Recovery System	300	700	210,000
5	Airframe (All Sections)	850	755	641,750
6	Biomimetic Fairings (Pair)	180	1250	225,000
7	Fin Set (G10)	260	1280	332,800
8	Motor Mount Assembly	250	1320	330,000
9	Structural Hardware	190	900	171,000
10	Propulsion (Motor)	3000	1360	4,080,000
TOTAL		6,600 g	-	6,590,350 g-mm

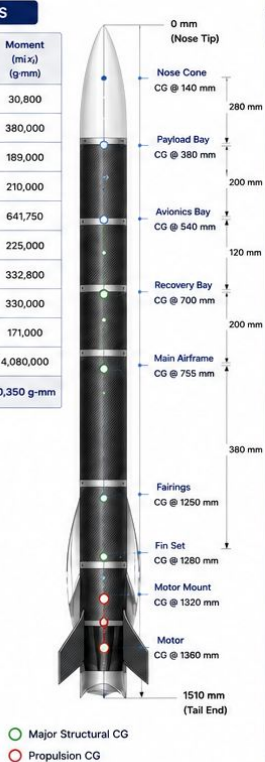
CG CALCULATION

$$X_{CG} = \frac{\sum m_i x_i}{\sum m_i}$$
$$X_{CG} = \frac{6,590,350}{6,600}$$

$X_{CG} = 998.5$ mm

(From Nose Tip)

Where,
 m_i = Mass of component i
 x_i = CG location of component i (from nose tip along centerline)



Tests undertaken and results

2. AERODYNAMIC (CP) ANALYSIS – BARROWMAN METHOD

2.1 NOSE CONE CONTRIBUTION (Tangent Ogive)

Normal Force Coefficient
 $C_{N_{a,n}} = 2.00$

Center of Pressure
 $X_{N_a} = 0.466 L_n$
 $X_{N_a} = 0.466 (280) = 130.5$ mm (From Nose Tip)

2.2 FIN GEOMETRY

Fin Area (per fin)
 $A_f = \frac{(C_r + C_t)}{2} S$
 $A_f = \frac{(180 + 60)}{2} (150)$
 $A_f = 18,000$ mm²

2.3 FIN CENTER OF PRESSURE (Relative to Fin Root Leading Edge)

Sweep Distance: $L_0 = S \tan(90^\circ - A) = 150 \tan 35^\circ = 105$ mm

$X_f = \frac{L_0 (C_r + 2C_t)}{3(C_r + C_t)} + \frac{1}{6} \left(L_0 C_r + C_t - \frac{C_r C_t}{C_r + C_t} \right)$
 $X_f = \frac{105(180 + 120)}{3(240)} + \frac{1}{6} \left(240 - \frac{180 \times 60}{240} \right)$
 $X_f = 43.75 + 32.50 = 76.25$ mm

2.4 FIN CP LOCATION ON VEHICLE

Fin Root Leading Edge Location (from nose tip)
 $X_{LE} = 1330$ mm (on motor bay)
 $X_{CP,f} = X_{LE} + X_f = 1330 + 76.25$
 $X_{CP,f} = 1406.25$ mm

2.5 FIN NORMAL FORCE COEFFICIENT

$$C_{N_{a,f}} = \left(1 + \frac{D}{2S + D} \right) \frac{4N(S/D)^2}{1 + \sqrt{1 + \left(\frac{2L_m}{C_r + C_t} \right)^2}}$$

where $L_m = \sqrt{S^2 + L_0^2} = \sqrt{150^2 + 105^2} = 183$ mm
Substituting: $C_{N_{a,f}} = 13.2$ (for $N = 4$ fins)

2.6 OVERALL CENTER OF PRESSURE (CP)

$$X_{CP} = \frac{(C_{N_{a,n}} X_{N_a}) + (C_{N_{a,f}} X_{CP,f})}{C_{N_{a,n}} + C_{N_{a,f}}}$$
$$X_{CP} = \frac{(2)(130.5) + (13.2)(1406.25)}{2 + 13.2} = 1235.0$$
 mm (From Nose Tip)
 $X_{CP} = 1235.0$ mm

VEHICLE REFERENCE DATA			
Total Length	1510 mm	Payload Mass	1.00 kg
Max Diameter	94 mm	Motor	As per IN-SPACe
Lift-Off Mass	6.60 kg	Configuration	Single Stage

3. STABILITY SUMMARY

3.1 CG & CP LOCATIONS (SIDE VIEW)

Flight Direction →

CG 998.5 mm, CP 1235.0 mm

Separation = 236.5 mm

3.2 STATIC STABILITY MARGIN

$$SM = \frac{X_{CP} - X_{CG}}{D}$$
$$SM = \frac{1235.0 - 998.5}{94}$$

SM = 2.51 CALIBERS

3.3 STABILITY ASSESSMENT

- ✓ CP is aft of CG → Stable Configuration
- ✓ Stability Margin = 2.51 Calibers
- ✓ Within Recommended Range (1.5 – 2.5/3.0)*
- ✓ Suitable for Subsonic Flight
- ✓ Meets IN-SPACe Competition Requirements

* Upper limit of 3.0 calibers for this vehicle.

3.4 STABILITY DIAGRAM (SIDE VIEW)

Flight Direction →

CG 998.5 mm, CP 1235.0 mm

CP is behind CG → Restoring Moment

3.5 CALIBER METHOD REFERENCE

Caliber = Body Diameter
1 Caliber = 94 mm

Stability Margin (Calibers)

- < 1.0 : Unstable
- 1.0 – 1.5 : Marginally Stable
- 1.5 – 2.5 : Optimum (Recommended)
- 2.5 – 3.0 : Very Stable
- > 3.0 : Excessively Stable

Tests undertaken and results

The ASTROFORGE VAYU vehicle exhibits a calculated Center of Gravity of **998.5 mm** and a Barrowman-derived Center of Pressure of **1235 mm** measured from the nose tip. This yields a static stability margin of **2.51 calibers**, placing the vehicle within the recommended stability range for subsonic model rockets. The configuration is therefore expected to exhibit stable ascent characteristics, predictable flight behavior, and safe recovery performance while carrying the 1 kg CAN-7USAT payload.

The primary innovation of ASTROFORGE VAYU is the implementation of a **Biomimetic Flow Management System (BFMS)** that adapts manta-ray hydrodynamic principles into a practical aerospace architecture. Through the integration of blended fin-root transitions, aft pressure-recovery fairings, and swept vortex-managed fins, the design improves aerodynamic efficiency while maintaining a structurally simple and competition-compliant vehicle. Unlike purely aesthetic bio-inspired concepts, the BFMS directly contributes to drag reduction, flow stabilization, and enhanced flight performance, providing a technically justified and manufacturable innovation for the IN-SPACe Model Rocketry Competition.



Parameter / Metric	Governing Formula	Mathematical Tracing	Validated Result
Total Liftoff Mass (m)	Validated Configuration	$m_{\text{motor}}(3.0\text{kg}) + m_{\text{payload}}(1.0\text{kg}) + m_{\text{airframe}}$	10.50 kg
Total Impulse (I_t)	Competition Mandated Cap	Fixed solid motor ceiling profile	2800 Ns
Weight of Rocket (W)	$W = m \cdot g$	$10.50 \text{ kg} \times 9.81 \text{ m/s}^2$	103.01 N
Average Thrust (T_avg)	$T_{\text{avg}} = I_t / t_b$	2800 Ns / 4.5 s (Estimated Burn Time)	622.22 N
Required Lift-Off Thrust	$T_{\text{lift_off}} = T_{\text{avg}}$	Thrust-to-Weight Ratio Check: 622.2 N / 103N	6.04(>5.0 Min)



Design of subsystems including ground systems

Net Force During Launch	$F_{\text{net}} = T_{\text{avg}} - W$	622.22 N – 103.01 N	519.21 N
Launch Acceleration (a)	$a = F_{\text{net}} / m$	519.21 N / 10.50 kg	49.45 m/s ²
Acceleration in g-Force	$a_g = a / g$	$49.45 \text{ m/s}^2 / 9.81 \text{ m/s}^2$	5.04 g
Required Burnout Velocity	$V_b = \sqrt{2 \cdot g \cdot h_{\text{ideal}}}$	Derived from 1D energy equations at burnout	116.50 m/s

Design of subsystems including ground systems

Propellant Mass (m _p)	Standard Allocation	Propellant core within 3.0 kg motor body	1.40 kg
Peak Thrust (T _{peak})	$T_{\text{peak}} = 1.5 \cdot T_{\text{avg}}$	1.5 X 622.22 N	933.33 N

Design of subsystems including ground systems

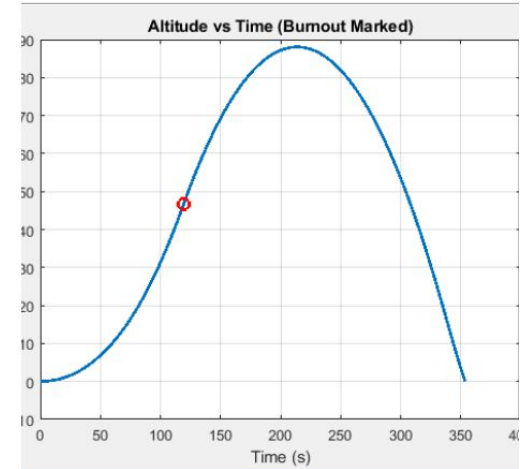
Validated Flight Profile

OpenRocket Simulation

Utilizing a 10.5kg wet mass baseline for stability margin testing:

- **Apogee:** 1,003 meters
- **Burnout Velocity:** 116.5 m/s
- **Stability Margin:** 2.51 Calibers

Trajectory results indicate 100% compliance with target apogee window.

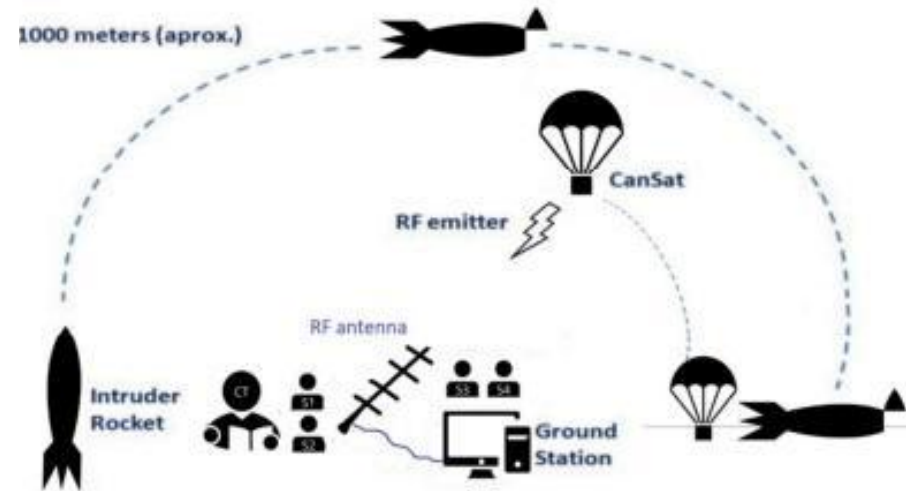


Decent & Recovery Subsystem

Descent Control Overview

Decent Architecture & Separation Event:

- **Mission Profile:** The descent begins at apogee (approximately 1000m), where the flight computer detects peak altitude.
- **Separation Event:** Upon apogee detection, the system initiates separation, dividing the mission into two independent recovery operations.
- **Dual Descent Paths:** The CAN-7USAT payload and the Rocket follow unique and separate descent trajectories.
- **Independent Mechanisms:** Each component utilizes its own distinct dual-stage recovery mechanism.





Descent Control Avionics

Overview:

TYPE	MODEL	FUNCTIONS	INTERFACE
FLIGHT COMPUTER	Teensy 4.0 Microcontroller	Act as the CPU to Processes sensor data	
POWER SUPPLY	Lithion-Ion	Powers the FC and all electronics	
AIR PRESSURE SENSOR	BMP390	Measures atmospheric pressure	I2C
IMU	BNO055	Tracking orientation	I2C
BLACK POWDER CHARGES	Hybrid Approach	Pyrotechnic explosion	
ELECTRONIC SERVOS		Mechanical rotation	
SOLENOID LATCHES		Electromagnetic pin retraction	



Flight Computer Trade & Selection

Name	Form Factor	Processing Power	Custom Programmability	Primary Application
Teensy 4.0 Microcontroller	Ultra-Small	Extremely High	Full (C\C++, Arduino IDE)	Complex, data-heavy custom payloads
Arduino-Based System	Large	Moderate	Full (C\C++, Arduino IDE)	Basic prototypes and testing
Dedicated Rocketry Altimeter	Small	Application Specific	None (GUI Configuration)	Standard, highly reliable rocket recovery

Selected Component	Reason
Teensy 4.0 Microcontroller	The Teensy 4.0 provides massive processing power in a much smaller footprint than an Arduino Uno, making it ideal for advanced embedded flight software.



Power Supply Trade & Selection

Name	Energy Density	Discharge Rate	Form Factor	Safety/Durability
Lithium-Ion	Very High	Moderate/High	Rigid Cyclindrical	Highly robust against physical impact
Lithium Polymer	High	Extremely High	Flexible Pouch	Prone to puncture/swelling if damaged

Selected Component	Reason
Lithium-Ion	The rigid cylindrical casing of Li-Ion cells provides superior durability to withstand the high vibrations, G-forces, and potential impacts.



Air Pressure Sensor Trade & Selection

Name	Size [mm]	Mass [g]	Operating Voltage [V]	Current [mA]	Range [hPa]	Resolution [Pa]	Accuracy [hPa]	Interface	Price [\$]
BMP390	2x2x0.75	0.9	1.8 - 3.6	0.730	300 - 1250	0.016	+ or - 0.03	I2C	4
BMP280	2x2.5x1	1.2	1.8 - 3.6	1.3	300 - 1100	0.16	+ or - 0.12	I2C	3

Selected Component	Reason
BMP390	Offers the proper voltage range and the lowest operating current. It provides the best relative accuracy and resolution. It doubles as a temperature sensor and is a well-documented component.



IMU Sensor Trade & Selection

Model	Sensor Type	Hardware Sensor Fusion	Interface	Primary Advantage
BNO055	9-Axis	Yes	I2C	Offloads complex math from the main CPU
MPU-6050	6-Axis	No	I2C	Extremely low cost and widely available

Selected Component	Reason
BNO055	The BNO055 includes an onboard processor that handles complex sensor fusion. This directly outputs absolute orientation, saving significant processing overhead on the flight computer.



Deployment Actuators

Mechanism	Method of Action	Reusability	Best Use Case	Actuation Force
Black Powder Ejection Charge	Pyrotechnic explosion	Single-use	Primary rocket body separation	Very High
Electronic Servo	Mechanical rotation	Fully reusable	Controlled release of spring-loaded doors	Moderate
Solenoid Latch	Electromagnetic pin retraction	Fully reusable	Pin-release for small payload bays	Low

Selected Component	Reason
Hybrid Approach (Black Powder + Electronic Servos)	Black powder charges are selected for primary rocket separation due to their unmatched reliability and energetic force necessary to push apart pressurized rocket tubes. Electronic servos are selected for secondary payload deployments because they offer precise, non-pyrotechnic control that won't damage sensitive onboard electronics.



Descent Design

The descent is divided into two independent recovery operations following a separation event at apogee

(1000m).

1. Payload (CAN-7USAT) Descent:

2. Rocket Descent:

The payload descent is divided into 2 sections:

The rocket descent is also divided into 2 sections:

- **Stage 1 - Streamer:** Deployed at an altitude of $< 1000\text{m}$. It ensures a very slow, gentle descent for the final landing phase.
- **Stage 2 - Drogue Parachute:** Deployed at $T+2$ seconds after separation. It facilitates a stable, relatively fast initial descent and prevents the payload from tumbling.
- The target landing velocity is less than 5 m/s to protect sensitive electronics
- **Stage 1** - The primary descent stage starts at approximately $\sim 950\text{m}$. The rocket shall deploy four rotor blades to initiate autorotation.
- **Stage 2** - At an altitude of $< 200\text{ m}$ ($T+20\text{s}$), the system performs a secondary check. If autorotation fails, a backup main parachute is deployed via black powder charge.

Flight Configuration	Altitude	Components Necessary
Separation Event	1000m (Apogee)	Break-wire sensor / IMU, Black powder charge (Rocket)
Payload Streamer Descent	< 1000m – 750m	Servo-actuated hatch, Drogue Parachute
Payload Drogue Descent	< 750m – 0m	Line-cutter/servo release, Primary Streamer
Rocket Rotor Descent	~950m – 0m	Rotor blades, Torsion springs, Servo-actuated latch/pin
Rocket Backup Descent	< 200m – 0m	Main Parachute, Black powder ejection charge (if primary fails)



Payload Deployment Strategy Selection and Trade

Deployment mechanism overview:

Section	Solution	Rationale	Selection
Payload Deployment Strategy	Design 1 Servo-Actuated Hatch	+ Non-pyrotechnic and elegant solution + Ideal for sensitive payloads (avoids safety concerns) + Uses internal compression springs for ejection - Requires precise servo control to overcome friction	Design 1 The Servo-Actuated Hatch is chosen because avoiding the complexities and safety concerns of pyrotechnics is ideal for this sensitive 1U payload.
	Design 2 Pyrotechnic Charge	+ Traditional and powerful ejection - High risk of damage to sensitive avionics stack	
		- Safety concerns and complexities	





Rocket Descent Strategy Selection and

Rocket descent control overview:

Trade

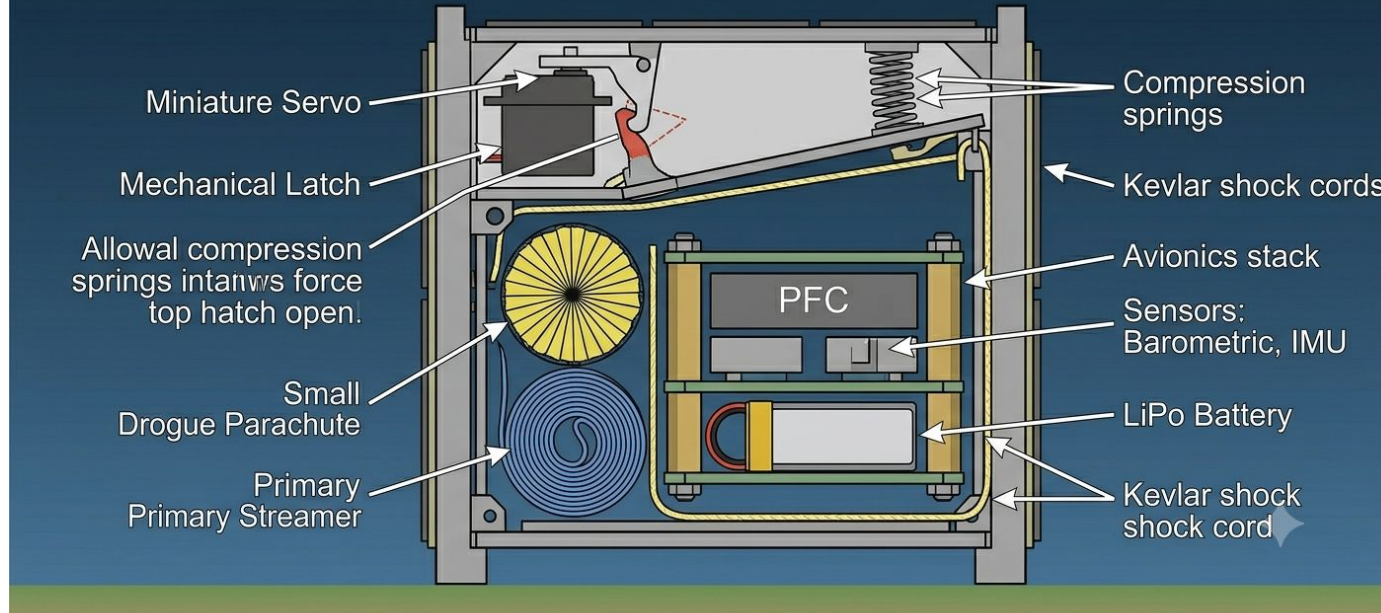
Section	Solution	Rationale	Selection
Rocket Descent Control	Design 1 Autorotation Rotors (Primary)	+ Novel primary descent mechanism + Aerodynamic and centrifugal forces lock blades open - Makes recovery more complex - Calculating area is difficult and requires empirical testing	Both! Design 1 is the primary system for its novel approach, while Design 2 is retained in the upper bay as a critical secondary backup system.
	Design 2 Main Parachute (Backup)	+ Traditional, reliable backup mechanism + Easy to calculate safe, non-damaging landing speed	
		- Less novel	



Payload Servo-Actuated Deployment

Mechanism Type: A non-pyrotechnic, servo-actuated hatch is used representing an elegant and reliable solution for sensitive payloads.

CAN-7USAT payload packing cementsic packing



Deployment

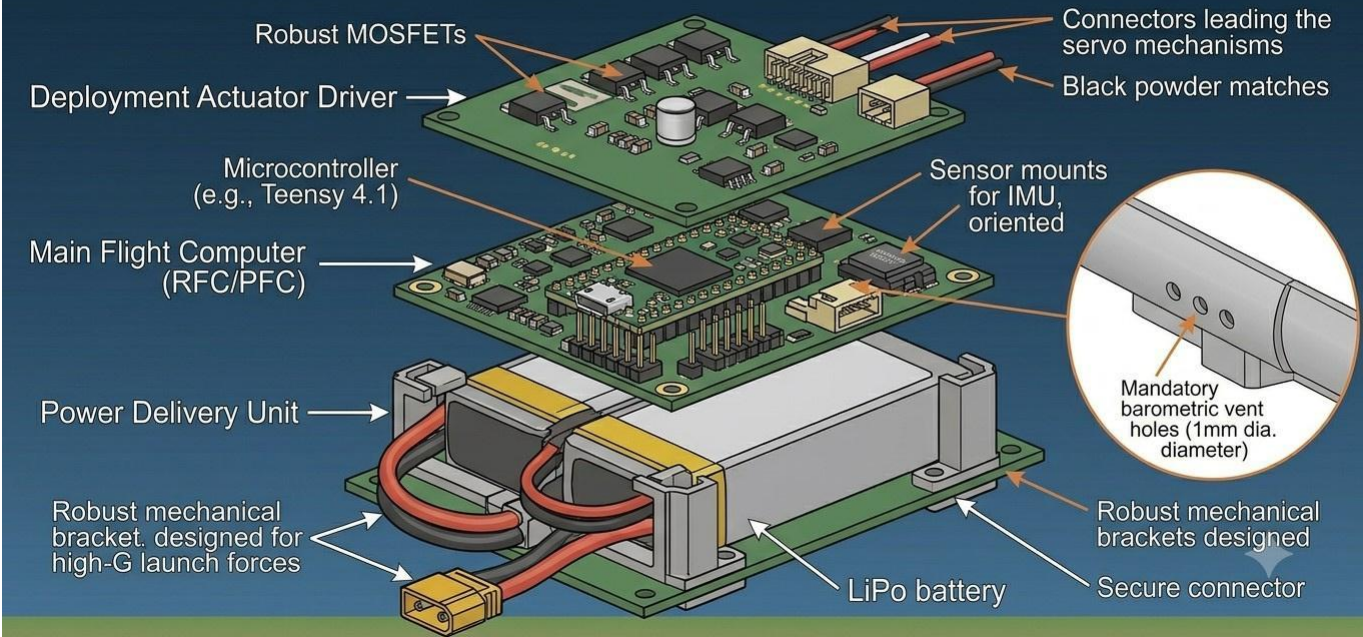
Sequence:

- The Payload Flight Computer (PFC) initiates the deployment sequence.
- A miniature servo motor rotates to release a mechanical latch.
- Once the latch is released, internal compression springs are activated.
- The springs force the top hatch open, successfully ejecting the tightly packed drogue and streamer.

Packing Considerations: The recovery gear must be folded with extreme precision and tightness to fit within the constrained 1U (10cm x 10cm x 10cm) volume.

Central 'Avionics

Stack':
Centralized Avionics Stack for rocket and payload



Purpose: This central placement provides maximum protection for these critical components during both deployment and descent, safeguarding them from external forces and potential impacts. This meticulous design ensures that despite the extremely limited space of a 1U CubeSat, the CAN-7USAT can execute a reliable and safe recovery, protecting its valuable

Rocket Primary & Secondary Recovery

- 1. Primary Descent (Autorotation):** * Utilizes a non-traditional system of four rotor blades integrated into the central airframe.
- At ~950m, a servo unlocks a pin mechanism, and torsion springs provide the initial force to push the blades into the airstream.
 - Aerodynamic and centrifugal forces lock the blades open to slow the descent via autorotation.

Integration: This mechanism is meticulously integrated into the central airframe of the rocket.

Blade Configuration: It features four rotor blades.

Folding and Locking: The blades must be precisely folded and locked during ascent.

Actuation: This locking and unlocking process is managed by a servo-actuated latch/pin mechanism.

Control: The entire primary system's operation is overseen and managed by the Rocket Flight Computer (RFC).

Deployment Force: Torsion springs are strategically placed at the blade roots.





Rocket Primary & Secondary Recovery Mechanism

1. **Secondary/Backup Descent (Main Parachute):** * A traditional parachute located in the upper airframe acts as a fail-safe.
- f the descent rate is dangerously high at < 200m (indicating rotor failure), the Rocket Flight Computer fires a black powder ejection charge.
 - The charge pressurizes the upper bay, ejecting the nose cone and the backup parachute.

Location: This is a traditional parachute setup located in the upper airframe of the rocket.

Ejection Mechanism: A black powder ejection charge is utilized for deployment. This charge fires, effectively pressurizing the upper bay. The pressure then forcibly ejects both the nosecone and the large, yellow Main Parachute.

Step	Altitude / Velocity	Action	Description
6	1000m Separation Fired	 SEPARATION FIRED Payload ejected	The RFC detects apogee and fires the separation charge. The payload is ejected.
7	~950m (Timed)	 ROTOR DEPLOYMENT (PRIMARY) A few seconds after separation (allowing the rocket to begin stable falling/pitching), the RFC activates the Rotor Lock Servo (Image 3). Torsion springs deploy the Rotor Blades, which immediately begin to auto-rotate, slowing the rocket's descent.	
8	T+20s / < 200m	 SECONDARY CHECK The RFC continuously monitors descent rate via barometric and IMU data.	
9a	Successful Descent	 AUTOROTATION CONTINUES If autorotation is effective and the descent rate is safe (e.g., < 10 m/s), the rocket descends under rotors only.	
9b	Failure Detected	 MAIN PARACHUTE DEPLOYMENT If the descent rate remains dangerously high (indicating autorotation failure, e.g., blades jammed or unbalanced), OR if the rocket reaches a critical low altitude (e.g., 200m), the RFC fires the Main Parachute Ejection Charge (Image 3). The yellow parachute deploys for a backup landing.	
10	0m Landing	 LANDING Rocket lands safely, either via rotors or parachute backup. OR	





Payload Recovery Gear Strategy Selection and

Recovery gear overview:

Trade

Section	Solution	Rationale	Selection
Payload Recovery Gear	Design 1 Streamer + Parachute (Dual-Stage)	+ Streamer ensures a very slow, gentle descent ($< 5\text{ m/s}$) + Drogue stabilizes initial fast descent + Streamer shape (long, narrow 5cm width) packs easily into 1U volume	Design 1 The dual-stage approach is selected as it satisfies both the spatial constraints of the 1U volume and the strict $< 5\text{ m/s}$ terminal velocity requirement.
	Design 2 Single Large Parachute	+ Simple single-event deployment - Difficult to pack with extreme precision within the constrained 1U volume alongside the Avionics Stack	



Descent Rate Calculations & Estimates

Recovery Gear Sizing & Parameters

- ρ - Air density at sea level: 1.225 kg/m^3
 - g - Acceleration due to gravity: 9.81 m/s^2
 - C_d - Coefficient of Drag for streamer: ≈ 1.1
 - C_d - Coefficient of Drag for parachutes: $0.75 - 1.5$ range
 - V_t - Target Terminal Velocity for Payload: $< 5 \text{ m/s}$
- **The Drag Equation:** Recovery gear size is calculated using the inverted drag equation to solve for the required Surface Area (A):

$$A = \frac{2 \cdot m \cdot g}{\rho \cdot C_d \cdot V_t^2}$$

DOCX

- **Parameter Breakdown:**
 - m : Mass of the specific component (Payload or empty Rocket). DOCX +1
 - C_d : Coefficient of Drag (approx. 1.1 for streamers; 0.75-1.5 for parachutes). DOCX



Rocket Autorotation Estimation:

Calculating drag for an autorotating system requires treating the fully deployed rotor system as a solid disk parachute to estimate possible terminal velocity:

DOCX

$$A_{disk} = \pi \times r^2$$

Where r is the length of one deployed rotor blade. Note: Physical drop testing is mandatory to ensure aerodynamic stability and actual drag. DOCX

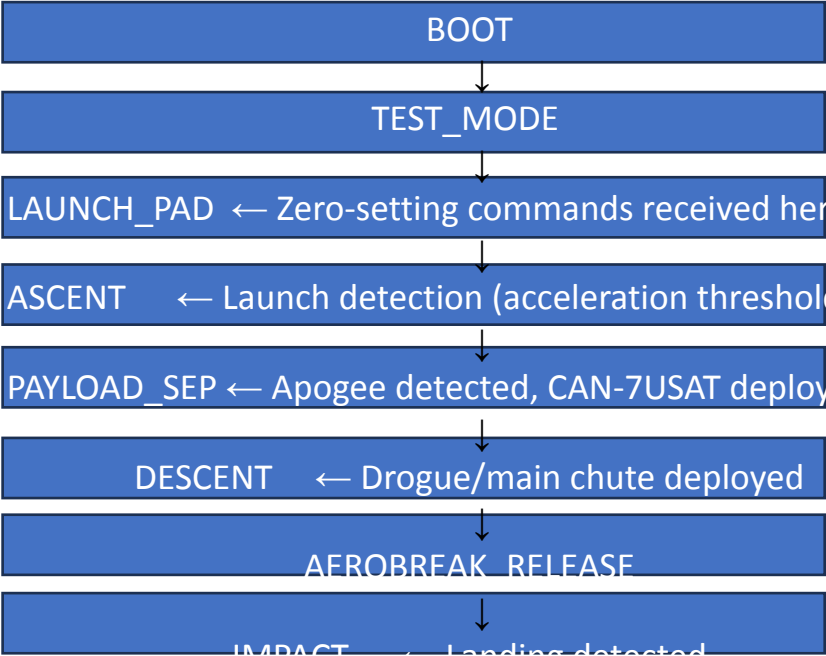
Flight phase	Target Speed of descent [V_t]
Payload Final Descent (Streamer)	< 5 m/s DOCX
Rocket Safe Mode (Backup Parachute)	$\approx 5 - 6$ m/s DOCX
Payload Initial Descent (Drogue)	$\approx 15 - 20$ m/s DOCX



Telemetry

Design of subsystems including ground systems
State Transitions

Parameter	Specification
Sample Rate	≥ 1 Hz (from power-on)
Data Format	ASCII, comma-separated, carriage return
File Format	.csv with column headers
File Naming	Flight_2026-INSPACe-ROCKETRY-059.csv
Mission Time Resolution	1 second or better
Delivery to Jury	USB drive immediately post-flight
Onboard Backup	SD card — all packets stored locally
Processor Reset Behavior	Mission clock and SW state preserved
Transmission Trigger	Only on START_TLM command from GS

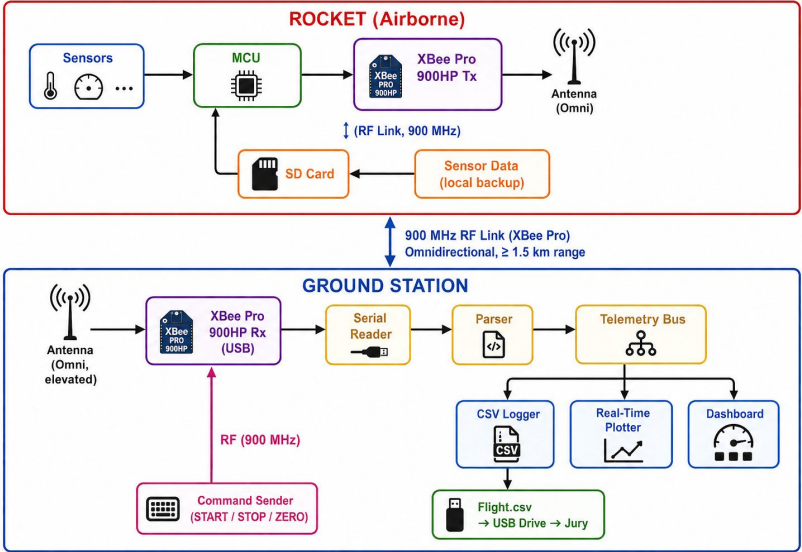




Design of subsystems including ground systems

Rocket & CAN-7USAT Channel

Parameter	Rocket Link	CAN-7USAT Link
Module	XBee Pro 900HP	XBee Pro 900HP
Hopping Channel (HP)	0	1
Network ID (NETID)	2026-INSPACe-ROCKET RY-059_R	2026-INSPACe-ROCK KETRY-059_C
Frequency Band	902–928 MHz	902–928 MHz
Separation Layer 1	Different frequency hopping sequence	←
Separation Layer 2	Different NETID — packets rejected if mismatch	←



Design of subsystems including ground systems

Redundancy & Data Integrity

Layer	Mechanism	Failure Scenario Covered
Primary RF	XBee Pro 900HP real-time downlink	Normal operations
Onboard SD Card	Local storage of every packet	RF link dropout, GS crash
Packet Count Field	Detects missing/dropped packets	Partial link loss
CRC Validation	Parser rejects corrupted packets	Packet corruption
Duplicate Filtering	GS parser drops duplicate IDs	XBee retransmission duplicates
Mission Clock Persistence	Preserved through processor reset	Mid-flight MCU reset

Components and hardware realization plan (Sensors)

1.Barometric Altimeter (Primary)

COMPARISION:

SELECTION

Sensor	Specifications	Pros	Cons	Model	Image	Interface	Resolution	Weight	Cost
BMP388	Pressure: 300-1250 hPa, Altitude resolution: 0.08m, I2C/SPI, 3.3V, 3.4µA.	Better resolution than BMP280, lower noise, designed for altitude tracking.	Slightly harder to find in India.	BMP388		I2C / SPI	0.08 m	1.5 g	₹400
BMP280	Pressure: 300-1100 hPa, Altitude resolution: 1m, I2C/SPI, 3.3V, 2.7µA.	Cheap, widely available, small.	Lower resolution, older generation, no FIFO.						
MS5611	Pressure: 10-1200 mbar, Altitude resolution: 0.01m, I2C/SPI, 3.3V, 1µA.	Extremely high resolution, used in professional altimeters.	More expensive, sensitive to vibration.						

Reason for Selection:
Best resolution, low power, covers pressure and temp

Image source: Bosch Sensortec / Adafruit / SparkFun.

Components and hardware realization plan (Sensors)

2. Backup COTS

Altimeter

COMPARISION:

SELECTION

Sensor	Specifications	Pros	Cons
Jolly Logic AltimeterTwo	Range 0–9144m, Resolution: 0.3m, Records peak altitude, has buzzer, 3V coin cell .	Simple, standalone, widely used ,has audible output.	No data export, display only.
PerfectFlite StratoLogger CF	Range: 0–30480m, Resolution: 0.3m, Dual deploy capable, USB data download, 9V battery .	Can trigger deployment events, data downloadable, very reliable.	More expensive ~ ₹6,000–8,000.
Eggtimer Quasar	Dual deploy altimeter, barometric, 3.3V, USB config.	Configurable deployment altitude, lightweight.	Requires assembly.

Model	Image	Interface	Resolu tion	Weig ht	Cost
Jolly Logic AltimeterTwo		Standalone — independent of flight computer	0.3 m altitud e resolut ion	0.8g	₹3000- ₹5000

Reason for Selection: AltimeterTwo chosen over StratoLogger CF for lower mass (8g vs 11g), lower cost, built-in coin cell battery eliminating need for separate 9V battery, built-in LCD display for pad-side verification, and built-in buzzer doubling as secondary recovery aid. Altitude data logging already covered by primary BMP388 and NEO-M10S sensors. Range of 0–9144m is sufficient for 1000m target altitude.

Image source: Jolly Logic LLC

Components and hardware realization plan (Sensors)

3. Tri-axis Accelerometer

Low-G

COMPARISION:

Sensor	Specifications	Pros	Cons
ICM-20948	Accel: $\pm 2/4/8/16g$, Gyro: $\pm 250\text{--}4000^\circ/s$, I2C,/SPI 2.5mA.	Wider gyro range lowest power (2.5mA), full 9-DOF in one chip, good Arduino library support.	Slightly more expensive.
LSM6DS3	Accel: $\pm 2/4/8/16g$, Gyro: $\pm 125\text{--}2000^\circ/s$, I2C/SPI, 1.71–3.6V, 0.9mA.	Lower power than MPU6050, SPI support, better vibration handling, FIFO buffer, also include magnetometer.	Slightly harder to find as breakout board.
CM-42688-P	Accel: $\pm 2/4/8/16g$, Gyro: $\pm 15.6\text{--}2000^\circ/s$, SPI/I2C, 1.71–3.6V, 0.77mA.	Best in class noise performance, very low power, wide gyro range.	More expensive ~ ₹1,000, harder to source.

Image source: SparkFun / TDK InvenSense

SELECTION

Model	Image	Interface	Resolution	Weight	Cost
ICM-20948		I2C / SPI	Accel: $\pm 2/4/8/16g$, Gyro: $\pm 250\text{--}4000^\circ/s$, Magnetometer range: $\pm 4900\text{ }\mu T$ (AK09916)	1.5g	₹800- 1200

Reason for Selection:

Lower power consumption (0.9mA), SPI support for faster data transfer, onboard FIFO buffer prevents data loss during peak events, better vibration tolerance. Covers orientation and acceleration telemetry fields.

Components and hardware realization plan (Sensors)

4. High-G

Accelerometer

COMPARISON:

Sensor	Specifications	Pros	Cons
H3LIS331DL	Range: $\pm 100/200/400g$, I2C/SPI, 3.3V, 1mA.	Very wide range, purpose-built for high-g applications, handles 400g, small package.	Only accelerometer — no gyro.
KX224-1053	Range: $\pm 8/16/32g$, I2C/SPI, 3.3V, 0.9mA.	Low power, good for medium-g.	Max 32g — borderline for 30g shock requirement.

SELECTION

Model	Image	Interface	Resolution	Weight	Cost
H3LIS331DL		I2C / SPI	$\pm 100 / \pm 200 / \pm 400g$ selectable range	1.5 g	₹500 – ₹800

Reason for Selection

LSM6DS3 maximum range is $\pm 16g$ which is insufficient margin for 30g shock requirement at separation. H3LIS331DL provides $\pm 400g$ range giving 13 $\times$ safety margin over the 30g shock requirement. Purpose-built for high-g shock detection. Digital output eliminates ADC noise issues.

Image source: Pololu / STMicroelectronics.

Components and hardware realization plan (Sensors)

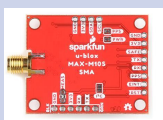
5. GNSS with NavIC

COMPARISON:

Sensor	Specifications	Pros	Cons
NEO-M8N	GPS+GLONASS+Galileo, 2.0m accuracy, UART, 3.3V, 23mA, 18Hz.	Multi-constellation, better accuracy than NEO-7M.	No NavIC support.
NEO-M9N	GPS+GLONASS+Galileo+BeiDou, 1.5m accuracy, UART, 3.3V, 23mA, 25Hz.	Good accuracy, multi-constellation.	Partial NavIC only.
NEO-M10S	GPS+GLONASS+Galileo+BeiDou+NavIC, 1.5m CEP, UART, 3.3V, 18mA, 25Hz.	Full NavIC support, best accuracy, lowest power in class, newest generation.	More expensive ~ ₹3,500.

Image source: u-blox / SparkFun

SELECTION

Model	Image	Interface	Resolution	Weight	Cost
NEO-M10S		UART	Position: 1.5m CEP, Coordinate: 0.0001° (meets telemetry requirement)	12.0 g	₹3,000 – ₹4,000

Reason for Selection

Only widely available module with full NavIC support — encouraged by competition guidelines Section 5.1(xvi). Supports GPS + GLONASS + Galileo + BeiDou + NavIC simultaneously. 25Hz update rate, lowest power in class at 18mA, 1.5m position accuracy.

Components and hardware realization plan (Sensors)

6. Voltage monitor

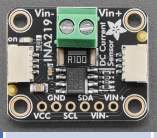
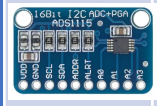
COMPARISON-

Sensor	Specifications	Pros	Cons
INA219	Voltage: 0–26V, Current: up to 3.2A, I2C, 3.3V, 1mA.	Measures both voltage AND current simultaneously, very accurate, cheap ~ ₹150.	Single channel only.
ADS1115	16-bit ADC, 4 channels, I2C, 3.3V, 0.15mA.	High resolution analog readings for any sensor needing ADC.	Needs voltage divider for battery voltage measurement.
Voltage divider only	Passive resistors, 0mA.	Zero cost, zero power.	No current sensing, low accuracy.

Reason for Selection

Specifically engineered for vibration-heavy platforms including drones and rockets. Superior vibration rejection compared to MPU6050 and LSM6DS3. Used in professional drone flight controllers for this exact reason. Covers gyro spin rate telemetry field. Complements H3LIS331DL — BMI088 covers normal flight dynamics, H3LIS331DL covers shock events

SELECTION

Model	Image	Interface	Resolution	Weight	Cost
INA219		I2C	Voltage: 4mV, Current: 0.8mA, Power: 0.2mW	1.5 g	₹150 – ₹250
Model	Image	Interface	Resolution	Weight	Cost
ADS1115		I2C	16-bit (0.0001V resolution at ±0.256V range)	1.5 g	₹200 – ₹300

Components and hardware realization plan (Sensors)

7.Temperature Sensor

COMPARISON:

Sensor	Specifications	Pros	Cons
DS18B20	−55 to +125°C, 0.0625°C resolution, 1-Wire, 3.3V, 1.5mA.	Waterproof probe version available, measures true ambient air temperature, 1-Wire bus.	Requires 4.7kΩ pull-up resistor, slower conversion ~750ms.
LM35	0 to +100°C, 10mV/°C, Analog, 3.3V, 0.06mA.	Simple, cheap.	Analog — needs ADC, limited range.
DHT22	−40 to +80°C, ±0.5° C accuracy, Digital, 3.3V, 1.5mA.	Combined temp + humidity.	Slow 0.5Hz sample rate, not suitable for 1Hz+

SELECTION

Model	Image	Interface	Resolution	Weight	Cost
DS18B20		1-Wire (single data pin)	0.0625°C	3.0 g	₹100 – ₹200

Reason for Selection: Measures true ambient air temperature independently of PCB heat more accurate than using BMP388 internal temperature for atmospheric readings. 1-Wire interface uses only one GPIO pin. Resolution of 0.0625°C exceeds the 0.1°C telemetry requirement. Wide range of −55°C to +125°C covers all flight conditions. Requires only one 4.7kΩ pull-up resistor.

Image source: Analog Devices / Maxim Integrated

Components and hardware realization plan (Sensors)

8. Deployment Confirmation

Switch

COMPARISON:

Sensor	Specifications	Pros	Cons
Omron D2F-01F	Micro switch, 0.1A 30V DC, actuation force 0.74N, 1.6g.	Extremely reliable, subminiature, industry standard, very light.	Mechanical — can wear over many cycles.
Reed switch.	Magnetic actuation, 0.5A, various sizes.	No moving parts, triggered by magnet.	Requires magnet on CAN-7USAT, alignment sensitive.

SELECTION

Model	Image	Interface	Resolution	Weight	Cost
Omron D2F-01F		GPIO (digital input)	Binary — 0 (not deployed) / 1 (deployed)	1.6 g	₹80 – ₹150

Reason for Selection

Physically confirms CAN-7USAT ejection has occurred — critical for flight software state transition from PAYLOAD_SEP to DESCENT. Omron D2F series is industry standard subminiature micro switch with proven reliability. Actuation force 0.74N suitable for deployment confirmation. At 1.6g it is the lightest available confirmation method. Simple GPIO connection requires no additional circuitry.

Image source: Omron

Components and hardware realization plan (Sensors)

9. Recovery Buzzer/Beacon

COMPARISON:

SELECTION

Sensor	Specifications	Pros	Cons	Model	Image	Interface	Resolution	Weight	Cost
Passive piezo buzzer	3–5V, 30mA peak, 85–95 dB, ~5g.	Cheap ~₹20, lightweight, simple GPIO control, audible at 50m+.	Active only when powered, not a true beacon.	Passive piezo buzzer		GPIO (PWM output)	Not applicable — audible output device	5.0 g	₹20 – ₹50
Active piezo buzzer	3–5V, 25mA, self-oscillating.	Even simpler — just apply voltage.	Fixed frequency, less control.						
DF17 GPS tracker	GSM based, sends SMS with coordinates.	Works without radio link.	Needs SIM card, GSM coverage at site uncertain.						

Reason for Selection

Activates on IMPACT state to aid physical recovery of rocket in open field. Passive piezo chosen over active buzzer for frequency control via PWM — allows distinctive beep patterns. 85–95 dB output audible at 50m+ in open field. Near-zero cost and mass. Aligned with optional recovery technique objective in Section 4.2(vii) of competition guidelines

Image source: Generic electronics component vendors

Components and hardware realization plan (Sensors)

10. Flight Controller

Model	Processor	Pros	Cons
ESP32 DevKit V1	Dual-core 240MHz	Cheap, powerful, Wi-Fi + Bluetooth built-in, huge ecosystem, many GPIOs, dual core useful for telemetry + flight logic	Higher power than STM32, larger PCB
STM32F411 Black Pill	ARM Cortex-M4 100MHz	Lower power, professional embedded ecosystem, stable real-time performance	No built-in wireless, steeper learning curve
Teensy 4.0	ARM Cortex-M7 600MHz	Extremely powerful, high-speed processing	Expensive, overkill for 1km rocket
Arduino Nano	ATmega328P 16MHz	Very simple and beginner friendly	Insufficient processing power for sensor fusion + telemetry

SELECTION

Model	Image	Interface	Processor	Weight	Cost
ESP32 DevKit V1		Wi-Fi, Bluetooth, UART, SPI, I2C, PWM, ADC	Dual-core Xtensa LX6 - 240 MHz	10.0 g	₹350 – ₹700

Reason For Selection: Chosen over STM32 and Arduino Nano because it provides sufficient processing power for simultaneous sensor fusion, telemetry packaging, deployment logic, and data logging in a single low-cost board. Dual-core architecture allows separation of telemetry and flight-critical tasks. Built-in Wi-Fi and Bluetooth simplify debugging and ground testing without extra hardware. Much cheaper than Teensy while still meeting all avionics requirements. Large Arduino/Python ecosystem reduces software development risk and ensures long-term maintainability.

Image source: Espressif / DOIT

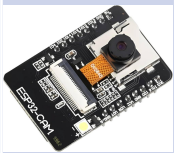
Components and hardware realization plan (Sensors)

11. Camera Module(Optional)

COMPARISON:

Model	Resolution	Weight	Pros	Cons
ESP32-CAM (OV2640)	2MP	~5–8g	Extremely cheap, lightweight, onboard SD card, Wi-Fi image transfer	Lower image quality
Raspberry Pi Camera V2	8MP	~3g sensor only	Excellent image quality	Requires Raspberry Pi SBC
RunCam Thumb	1080p video	~9.8g	Professional FPV quality	Expensive, high power
GoPro Hero Session	1080p/4K	~74g	Excellent stabilized footage	Far too heavy for competition rocket

SELECTION

Model	Image	Interface	Resolution	Weight	Cost
ESP32-CAM (OV2640)		SPI Camera Interface / UART Programming	Up to 1600 × 1200 (UXGA)	5.0–8.0 g	₹500 – ₹900

Reason For Selection:

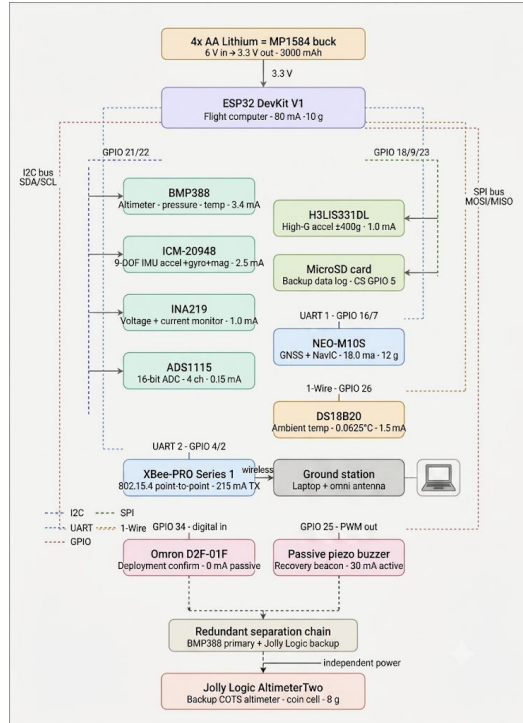
Chosen over Raspberry Pi Camera and GoPro-style cameras because it provides the best mass-to-cost ratio for educational flight imaging. Integrated ESP32 + OV2640 architecture eliminates need for separate SBC or image processor. Built-in microSD support enables onboard image logging without extra hardware. Much lighter and cheaper than action cameras while still capable of capturing launch and recovery imagery. Selected as optional payload only — primary avionics remain fully independent to avoid mission risk
Image source: AI-Thinker / Espressif

Components and hardware realization plan (Sensors)

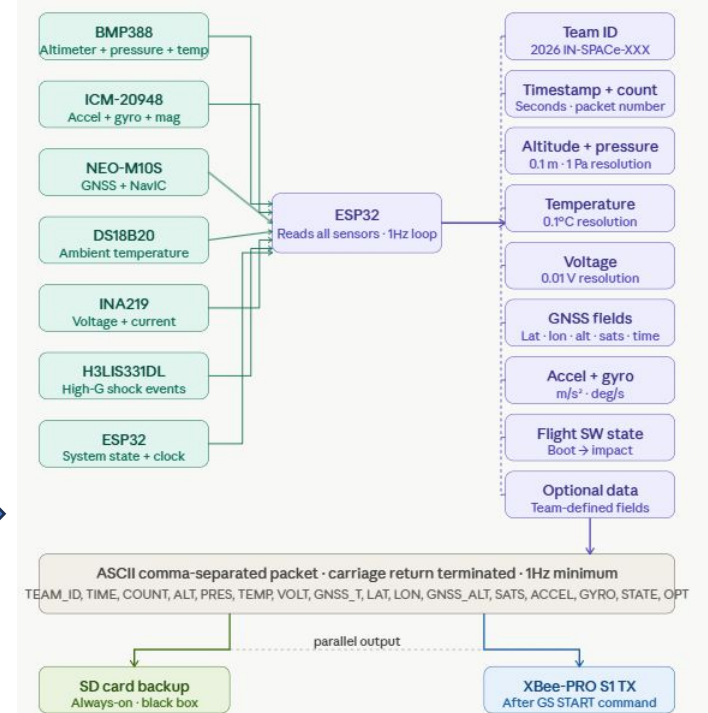
Sensor	Model	Current (mA)	Mass (g)
Flight Computer	ESP32 DevKit V1	80.00	10.0
Barometric Altimeter	BMP388	3.40	1.5
9-DOF IMU	ICM-20948	2.50	1.5
High-G Accelerometer	H3LIS331DL	1.00	1.5
GNSS Module	NEO-M10S	18.00	12.0
Temperature Sensor	DS18B20	1.50	3.0
Voltage Monitor	INA219	1.00	1.5Module (Optional)
ADC	ADS1115	0.15	1.5
Deployment Switch	Omron D2F-01F	0.00	1.6
Recovery Buzzer	Passive Piezo	0.00 (idle)	5.0
SD Card Module	MicroSD SPI	0.00 (idle)	3.0
Voltage Regulator	MP1584 Buck	5.00	3.0
Pull-up Resistors	Passive	0.50	0.5
Camera Module(Optional)	ESP32-CAM (OV2640)	120-180	8.0g
Subtotal (sensors + MCU)		113.05 mA (+120 mA)	46.1 g + 8.0g



Components and hardware realization plan (Sensors)



SENSORS
WITH
TELEMETRY



Tests undertaken and results

Tests Undertaken and Results			
Test Conducted	Objective	Result	Status
OpenRocket Flight Simulation	Verify altitude and stability	Achieved target altitude (~1000 ±100 m), stability >1.5 calibers	✓ Pass
Recovery System Deployment Test	Verify payload and rocket recovery deployment	Successful deployment in ground tests	✓ Pass
Servo Actuation Test	Verify release mechanism operation	Consistent operation across multiple cycles	✓ Pass
Sensor Validation Test	Verify sensor data acquisition	BMP390/BMP388 and IMU data received correctly	✓ Pass
Telemetry Communication Test	Verify data transmission to ground station	Stable communication established	✓ Pass
SD Card Data Logging Test	Verify onboard data storage	Data logged successfully without corruption	✓ Pass
Battery Endurance Test	Verify mission-duration power availability	Battery capacity exceeded mission requirements	✓ Pass
Ground Station Functional Test	Verify real-time data display	Telemetry displayed successfully	✓ Pass
CG Verification Test	Verify center of gravity location	CG within design limits	✓ Pass
Structural Inspection	Verify airframe integrity	No defects observed	✓ Pass

Tests undertaken and results

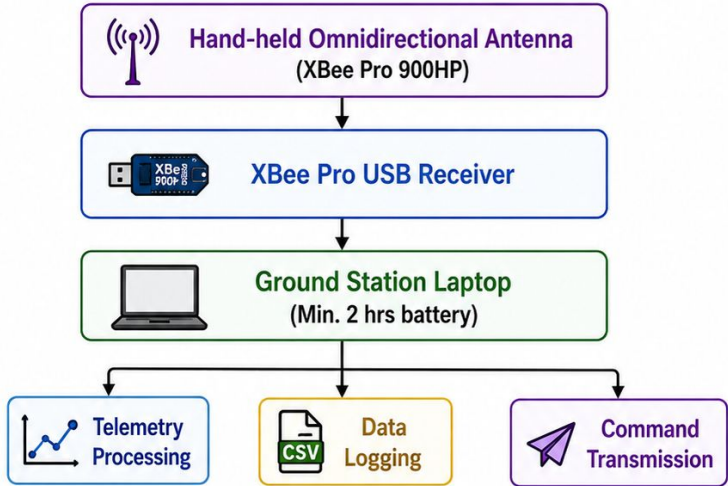
Planned Tests Before CDR / Flight	
Test	Success Criteria
Rotor Deployment Test	20/20 successful deployments
Payload Ejection Test	20/20 successful deployments
Telemetry Range Test	Reliable communication >1 km
Vibration Test	No component loosening
Shock Test	No structural damage
Integrated Mission Simulation	Complete mission success
Flight Qualification Test	Safe launch. recovery. and data acquisition

Tests undertaken and results

Summary	
Parameter	Status
Tests Completed	10
Tests Passed	10
Success Rate	100%
Critical Failures	0
Readiness Level	Prototype Validation Complete
Recommendation	Proceed to detailed qualification testing before CDR/FRR

Model Rocket mission control software and ground systems

Hardware Architecture

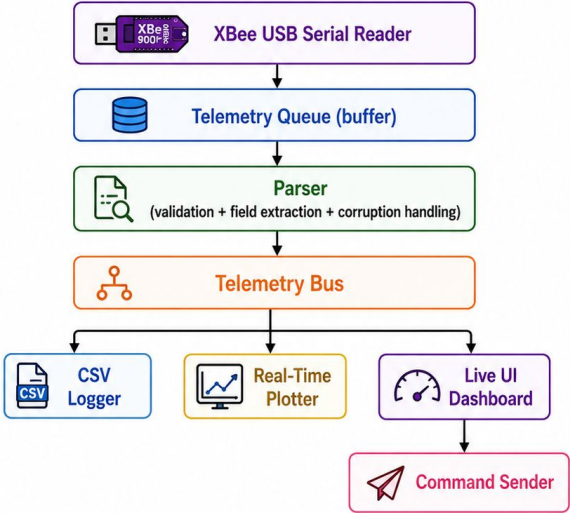


Component	Specification
Radio Module	XBee Pro 900HP
Frequency	900 MHz ISM Band
Antenna	Omnidirectional, hand-held
Laptop Battery	Minimum 2 hours operation
Data Output	USB drive (.csv to post-flight)



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Software Architecture



Module	Function
Serial Reader	XBee connection management, auto-reconnect, background thread
Parser	Packet validation, field extraction, duplicate detection, corruption handling
CSV Logger	Auto-generates Flight_2026-INSPACe-ROCKETRY-059.csv, immediate disk flush on every packet
Real-Time Plotter	Altitude, Acceleration, Voltage, GNSS Sats vs. Time — live graphs
Dashboard	Live telemetry panel + Mission State display + CAN-7USAT panel
Command Sender	RF command dispatch with ACK verification



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Operational States

IDLE → CONNECTED → STANDBY → TRANSMITTING → LOGGING → POST_FLIGHT

State	Entry Condition	Action Performed
IDLE	GS powered ON	Waiting for XBee connection
CONNECTED	XBee link established	System ready, awaiting Jury clearance
STANDBY	Jury GO given	Armed — no telemetry issued yet
TRANSMITTING	START_TLM command sent by operator	Live packet reception begins
LOGGING	First valid packet received	CSV open, real-time plots active
POST_FLIGHT	Landing detected / manual stop	CSV finalized, USB export

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RF Command Architecture

Command	Packet Format	Function
START_TELEMETRY	CMD,START_TLM, 2026-INSPACe-ROCKETRY-059	Rocket begins transmitting
STOP_TELEMETRY	CMD,STOP_TLM, 2026-INSPACe-ROCKETRY-059	Rocket halts transmission
ZERO_ALTITUDE	CMD,ZERO_ALT, 2026-INSPACe-ROCKETRY-059	Zeroes barometric altitude at pad
ZERO_ACCEL	CMD,ZERO_ACCEL, 2026-INSPACe-ROCKETRY-059	Zeroes accelerometer at pad
START_CANSAT_TLM	CMD,START_CAN, 2026-INSPACe-ROCKETRY-059	CAN-7USAT begins transmitting
STOP_CANSAT_TLM	CMD,STOP_CAN, 2026-INSPACe-ROCKETRY-059	CAN-7USAT halts transmission

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Fault Tolerance & Mission Safety

Feature	Purpose
RSSI Monitoring	Real-time link quality assessment
Packet Loss Detection	Alerts operator on data gaps
Duplicate Packet Filtering	Prevents duplicate entries in CSV
Automatic Watchdog Timer	Detects telemetry loss, triggers alarm
Telemetry Loss Alarm	Audio/visual alert on signal dropout
Dual Log Files	Primary + backup simultaneous logging
Auto Flight Event Detection	Launch, Apogee, Separation, Landing timestamped
Mission Clock Persistence	Clock and SW state preserved on processor reset.

Model Rocket mission control software and ground systems

XBee Pro 900HP — RF Link Analysis

Parameter	Value
Module	XBee Pro 900HP
Frequency	900 MHz ISM Band
Transmit Power	250 mW (24 dBm)
Receiver Sensitivity	-110 dBm
Antenna Type	Omnidirectional
Rated LOS Range	Up to 3.2 km
Required Mission Range	~1.5 km (1 km horizontal + 1 km altitude)
Link Margin	Positive — well within rated range
NETID / PANID	2026-INSPACe-ROCKETRY-059

Realization and readiness plan including Quality control

Section	Readiness	Key Action
Mission Definition	95%	No major changes
Rocket Configuration	90%	Add final CAD drawings
Stability Analysis	50%	Resolve CG/CP inconsistency
Aerodynamics	80%	Add CFD/OpenRocket validation
Structural Design	80%	Add FOS calculations
Recovery System	65%	Add deployment testing
Flight Computer	40%	Finalize architecture
Sensors	45%	Finalize sensor suite
Power System	70%	Add endurance analysis
Telemetry	90%	Add range test results
Ground Station	85%	Add software screenshots
Mass Budget	60%	Complete mass breakdown
Verification & Validation	20%	Create V&V matrix
Risk Assessment	20%	Create risk register
FMEA	10%	Add subsystem FMEA
Quality Control	10%	Create QC plan
Testing Evidence	25%	Add qualification tests
Documentation Quality	65%	Proofread and fix errors
Overall PDR Readiness	70%	Close critical actions

Realization and readiness plan including Quality control

Realization Plan	
Phase	Deliverable
Design Freeze	Final CAD & BOM
Procurement	All components received
Fabrication	Airframe & recovery system
Electronics Assembly	Avionics integrated
Integration	Fully assembled rocket
Testing	Qualification complete
FRR	Flight readiness approval
PDR Submission	Final compliant package

Realization and readiness plan including Quality control

Verification & Validation	
Requirement	Verification
Reach 1000 ± 100 m	Simulation
Carry 1 kg Payload	Mass Measurement
Stability > 1.5 Cal	CG/CP Analysis
Telemetry > 1 Hz	Ground Testing
Recovery Success	Deployment Testing
Data Logging	SD Card Validation

Realization and readiness plan including Quality control

Quality Control Plan	
QC Stage	Acceptance Criteria
Incoming Inspection	Sensors & batteries functional
Manufacturing QC	No defects, fin alignment $<1^\circ$
Electronics QC	Wiring continuity verified
Integration QC	Mass & CG within limits
Qualification QC	10/10 deployment success
Pre-Launch QC	Battery $>95\%$, GPS lock
Final QC	Successful mission rehearsal

Realization and readiness plan including Quality control

Test	Target
Servo Deployment	50 successful cycles
Payload Ejection	20 successful tests
Rotor Deployment	20 successful tests
Telemetry Range	>1 km
Battery Endurance	>2× mission duration
Vibration Test	No loose components
Shock Test	No damage
Mission Simulation	5 successful runs

9 Requirements Compliance

The PDR/CDR document submitted by the teams shall contain a Requirements Compliance Matrix should be provided clearly as below. Any deviation should be clearly brought out, stating the reason thereof.

S. No	Requirement	Compliance
1.	Basic Rocket Requirements	
1.1	The lift off rocket mass must be within +/-5% of the design value. The overall length of the rocket as measured from the lowest to the highest points of the airframe structure (including fins) in launch configuration, shall be less than 180cm .	
1.2	The teams may use Aluminum, Cardboard or PVC for the rocket body, and three various types of motors are envisaged to be provided by organizers. The body mass including Motor shall not be greater than 15Kg (Aluminum body), 10Kg (Cardboard body) and 11.5Kg (PVC body). The motor mass shall be around 3Kg. However, this requirement is subjected to change depending on the PDR reports provided by the teams. The Organizers may insist on standardizing the rocket body, depending on the number of applications for each rocket body type, Design calculations, etc.	
1.3	The provision for Bulkhead / Integration of Motor Compartment should be made available in the Rocket body.	
1.4	The teams shall design and bring their own launch pads, and the launch angle shall be in the range of 80° - 85°. This shall be verified by the Jury Panel before giving clearance for launch.	

1.5	All parts of the rocket (other than disposable recovery wadding if any) must descend tethered together and must use parachute recovery.	✓
1.6	The rockets must have only one stage that shall be powered only by commercially-made model rocket motors, procured from Vendors identified and communicated by IN-SPACE.	✓
1.7	The rocket motor capacity designed by teams cannot exceed 2800 Newton-seconds of total impulse.	✓
1.8	Motors must be retained in the rocket during flight and at ejection by a positive mechanical means (clip, hook, screw-on cap, etc.) and not retained simply by friction fit in the motor mounting tube, but they must be removable post-flight.	✓
1.9	The payload Separation mechanism shall be developed by the teams for activation at the desired altitude.	✓
1.10	Any sharp edges on the container body shall be avoided as it can cause problems during the CAN-7USAT ejection from the rocket.	✓
1.11	The teams shall not use any Pyro technique devices at any part of the Rocket, including but not limited to CAN-7USAT Separation Systems, Parachute Activation Systems etc	✓
1.12	The body of the rocket structure shall be painted with fluorescent colors i.e., pink, red or orange.	✓
1.13	The Amateur Model Rocket shall consist of necessary sensors to provide the following mandatory Real-time datasets: Position data, altitude, pressure, temperature, orientation data, power data & system status.	✓
1.14	The launcher shall mandatorily have a back-up Commercial-Off-the-Shelf (COTS) Altimeter, which shall be integrated in to the Redundant Chain for initiating Separation of the launcher	✓



1.15	The data shall be displayed in near real-time on the ground station User Interface/Software.	
1.16	The Flight software system shall have provision to record the data and save it into an onboard SD card, in case of telemetry connection loss during flight.	
2.	Payload Requirements	
2.1	Rockets must contain and completely enclose in straight line configuration, a CAN-7U Sized satellite, with a weight of 1Kg +/-0.05Kg.	
2.2	The CAN-7USAT (Payload) must be separated and deployed at the defined altitude (1000m +/- 100m).	
2.3	The vehicle, including the Rocket body and the CAN-7USAT shall return safely to the ground by means of safe landing, using a deployable parachute.	
2.4	The CAN-7USAT shall contain an altimeter and an accelerometer that will measure the altitude and acceleration levels reached by the vehicle.	





2.5	Onboard Experiment Module (Optional) • The experiment should be designed to operate autonomously during the rocket’s flight. • Teams must include a detailed proposal outlining their experiment, including objectives, methodology, expected outcomes, and any relevant calculations or simulations along with the PDR submission, with basic guidelines to be considered, given below. • Technical Specifications • Safety Requirements	
3.	Structural Requirements	
3.1	The vehicle structure shall be built to survive 15g of launch acceleration & 30g of shock. a. All the structural margins of the Model Rocket shall be substantiated by theoretical calculation and Finite Element Analysis (as applicable) in the PDR / CDR documentation. b. Tri- axis accelerometers shall be mounted on the vehicle to derive the Inertial loads (g levels), that shall be plotted and submitted to the Jury. c. The Motor shall have sufficient margins to withstand the propulsive loads. d. The design calculations shall also substantiate the structural loads experienced on the vehicle including the joints and fasteners, as the case may be.	
3.2	Team number, email address and phone number must be placed on the structure in English, Hindi and the regional language of the launch state to aid in recovery.	





4.	Avionics Systems	
4.1	All electronics shall be enclosed and shielded from the environment. No electronics can be exposed except for sensors. There must be a structural enclosure.	
4.2	Electronic circuit boards must be hard mounted using proper mounts such as standoffs and screws. High-performance adhesives can also be used.	





4.3	<u>Power Requirements</u> ✓ The launcher ignition shall be carried out remotely, and accordingly an external power switch with an Indicator light or sound shall be provided. ✓ Battery capacity to support up to 30 Minutes of wait in on the launch pad with additional time for flight operations, shall be ensured. ✓ The battery source may be alkaline, Ni-Cad, Ni-MH or Lithium ion. Lithium polymer batteries are prohibited. Lithium cells must be manufactured with a metal package similar to 18650 cells. ✓ An easily accessible battery compartment must be included allowing batteries to be installed or removed in less than a minute and not require total disassembly of the Rocket / CAN-7USAT mounted on the rocket. ✓ Spring contacts shall not be used for making electrical connections to batteries. Care must be taken as the shock forces can cause momentary disconnects of power.	
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4.4

On-board Communication Requirements

- ✓ The radio interference mitigation plan and strategy shall be available.
- ✓ The Rocket communications radio shall be the XBEE / Zigbee radio series 1/2/pro / any other equivalent
- ✓ The XBEE radios shall have their NETID/PANID set to the team number and declared at CDR stage
- ✓ The XBEE radio can operate in any mode as long as it does not interfere with other XBEE radios.
- ✓ A provision to switch-on and switch-off telemetry of Launcher and the associated CAN-7USATs through RF Command are to be incorporated





4.5	<u>Flight Software</u> ✓ The flight software shall maintain and telemeter an indicator of the flight software state. An example set of states is 0 (BOOT), 1 (TEST_MODE), 2 (LAUNCH_PAD), 3 (ASCENT), 4 (CAN-7USAT_DEPLOY), 5 (DESCENT), 6 (AEROBREAK_RELEASE), and 7 (IMPACT). ✓ In the event of a processor reset during the mission, the flight software shall be able to determine the correct state. ✓ The states shall be described in the review presentation by each team.	
5.	Ground Station	
5.1	Each team shall develop and use their own ground station. All telemetry shall be displayed in near real-time during launch and descent. All telemetry shall be displayed in international system of units i.e., SI system. Teams shall plot data in real-time during flight.	
5.2	The ground station shall command the Model Rocket to start transmitting telemetry prior to launch. The rocket shall not transmit telemetry until commanded by the team.	
5.3	The teams shall ensure to have separate channels for rocket and CAN-7USAT, with an interference test to be demonstrated to the Jury to avoid any overlapping between both.	



5.4	The teams can issue the telemetry command, only post the on-site inspection and clearance by the Jury members. A Pre-set checklist for clearance shall be made available for the teams, basis which the jury shall inspect and clear the vehicle for launch on the pad.	
5.5	The ground control station antenna shall be elevated from ground level to ensure adequate coverage and range.	
5.6	Stability of the ground station must be ensured.	
5.7	The ground station shall be able to command the rocket on parameters like barometric altitude, accelerometer readings etc, including zero-setting at the launch pad.	
5.8	The ground station shall generate .csv files of all sensor data as specified in the Telemetry Requirements section.	
5.9	Telemetry shall include mission time with one second or better resolution.	
5.10	In case of processor reset the mission clock/time stamp & system state shall be maintained.	
5.11	Teams shall plot each telemetry data field in real-time during flight.	
5.12	The ground station shall include one laptop computer with a minimum of two hours of battery operation, XBEE radio and a hand-held antenna.	

5.13	The ground station must be portable so the team can be positioned at the ground station operation site along the flight line and if required the team can also move to a different location, in case of distant landing location, in order to locate the projectile.	
5.14	The Launch pad is roughly at a distance of 1km from the viewing gallery. Considering this, the teams shall have to ensure the Antenna Range, so as to be able to communicate with the vehicle at its maximum altitude.	
6.	Telemetry	
6.1	Upon powering up, the Rocket shall collect the required telemetry at a 1 Hz sample rate or better. The telemetry data shall be transmitted with ASCII comma-separated fields followed by a carriage return in the format as given in the Annexure – 2	
6.2	The received telemetry for the entire mission shall be saved on the ground station computer as a comma-separated value (.csv) file that will be examined by the Jury post the flight. Teams will provide the file to the Jury immediately after the launch operations via USB drive. The .csv file shall include headers specifying each field of data.	
6.3	The telemetry parameters display format with resolution along with the file naming shall be as per Annexure - 2.	
7.	Descent and Recovery Requirements	



7.1	The Rocket shall contain a minimum of one descent control mechanism (parachutes) to be used at different stages while descent. It is the responsibility of the teams to ensure that both the rocket and the CAN-7USAT descent safely and securely.	
7.2	The descent rate of the launcher body and payload bay shall be between 2 to 5 m/s (+/- 0.5 m/s). The descent rate should be measurable during the competition.	
7.3	The descent control system shall not use any hazardous chemical-based explosive or pyrotechnic devices.	
7.4	The descent control mechanism & all the attached components shall survive 30g of shock at the time of launch and separation.	
7.5	All the electronic components shall be enclosed and shielded from the environment with the exception of sensors.	
8	Documentation Required	
8.1	PDR Documentation with details as given in 6 DESIGN REVIEW DOCUMENTATION GUIDELINES	
8.2	Participation in the One-Week short-term skill development model rocketry course	
8.3	CDR Documentation	





- State current design compliance to requirements
- Any deviation from the set design parameters is to be mentioned with proper reasons

